

Point-by-point Response to Reviewers' Comments:

Dear editor and reviewers:

We would like to thank the editor and reviewers for their encouraging, helpful and thoughtful comments. We feel that responding to these questions has made the paper considerably clearer and stronger. Our detailed responses to the reviewers' comments can be found below (reviewers' comments are italic). All changes to the revised manuscript were made in blue font.

Reviewer: 1

Summary

This prospective study of 56 patients with major depressive disorder evaluates macro- versus microscale hippocampal segmentation as a predictive imaging biomarker for antidepressant therapeutic response.

Comments to Authors:

***R1.1:** Please provide more detailed elaboration on findings that appear contrary to established literature. Both in the background and discussion sections, the study reports results that oppose much of the published literature, indicating that larger pre-treatment hippocampal volumes are associated with better treatment outcomes. A more thorough discussion of potential explanations for this discrepancy, including methodological differences, population characteristics, or alternative biological mechanisms, would strengthen the manuscript.*

Response: We thank the Reviewer for this insightful suggestion. We agree that the directionality of our findings—where smaller pretreatment volumes were associated with better response warrants careful interpretation given its divergence from some existing literature. In revised manuscript, we have discussed potential explanations for this discrepancy as suggested, including:

-methodological differences:

- **Population Characteristics:** We clarified that our findings in an early-stage, medication-free cohort represent a different phase of the disease compared to the chronic atrophy typically seen in multi-decade MDD cases (Page 20, Lines 53-59).

- **Biological Mechanisms:** We have more explicitly linked the "subtle volumetric excess" in non-responders to active inflammatory processes (e.g., edema) or pruning deficits, as opposed to the "neurogenic exhaustion" observed in late-stage illness (Page 21, Lines 15-24).

- **Methodological difference:** We have incorporated a remark in the Discussion to note that methodological differences may further contribute to these discrepancies. Specifically, our study utilized the updated FreeSurfer v7.1.1 (compared to v6.0 used in prior works), and employed different clinical outcome criteria and statistical modeling choices.

We believe these revisions better guide the reader through the rationale for our divergent findings.

R1.2: Did the authors consider the potential influence of illness duration in their analyses? This may affect hippocampal morphology and could potentially confound the relationship between

hippocampal volumes and treatment response. Please clarify whether this factor was examined and, if not, address this as a limitation.

Response: We thank the Reviewer for this important suggestion. We agree that illness duration is a potential confounder that could influence both hippocampal structure and clinical response. In initial submitted manuscript, we reported no significant difference in illness duration between the NRES and RES groups ($p=0.831$). This suggests that the stage of illness was well-balanced across different clinical outcomes.

To address the Reviewer's concern, we performed an additional ANCOVA to evaluate whether the predictive value of hippocampal metrics remained significant after adjusting for illness duration. We re-ran our primary models including illness duration as an additional covariate alongside age, sex, and ICV. As shown in the table below, the inclusion of illness duration had a negligible impact on the results; the predictive power of our core markers—the left posterior hippocampus (body and tail)—remained highly significant ($P<0.01$)

Table R1. ANCOVA Results With and Without Illness Duration Adjustment

Hippocampal Metrics	Primary Model (Age, Sex, ICV)		Extended Model (+ Illness Duration)	
	F	P	F	P
L-HE	8.642	0.031	8.380	0.033
L-HF	10.858	0.025	10.563	0.029
L-HP	7.128	0.031	6.941	0.033
L-Comb.Dentate	5.867	0.038	5.770	0.040
L-CA.Only	6.762	0.031	6.570	0.033
L-Comb.Dentate.CA	7.041	0.031	6.866	0.033
L-CA2.4	6.563	0.031	6.436	0.033
L-body	9.330	0.005	9.122	0.005
L-Hippocampal_tail	8.796	0.005	8.740	0.005
AI-HE	8.324	0.029	8.065	0.033
AI-HF	8.313	0.029	8.136	0.033
AI-body	5.840	0.057	5.636	0.048
AI-tail	5.096	0.057	5.566	0.048

Note: Results are based on the MDD cohort after excluding 4 extreme outliers in illness duration to ensure model stability ($N=52$).

Accordingly, we now clarify these findings into the Results section.

R1.3: *The authors conclude that macroscale segmentation outperforms microscale approaches, and in the limitations section, suggest future studies using ultra-high-field imaging (e.g., 7T MRI). However, did the authors consider utilizing high-resolution T2-weighted sequences on the existing 3T scanner? Such sequences are recognized as superior for hippocampal subfield delineation and may have enhanced the microscale analysis without requiring specialized equipment. Please comment on whether this approach was considered and, if so, why it was not implemented.*

Response: We thank the Reviewer for this suggestion. This study is a secondary analysis of a pre-existing research cohort, where the acquisition protocol was previously established. The

protocol focused on whole-brain T1-weighted imaging for broad neuroimaging research utility, while specialized hippocampal T2-weighted sequences were not included to limit scan time and ensure patient comfort. Although T2-weighted imaging is superior for subfield delineation, our use of standard 3D T1-weighted data enhances the clinical translatability of our findings, as T1 sequences are more widely accessible in psychiatric practice. We have added this context to the Limitations section.

Reviewer: 2

Comments to Author

Introduction:

R2.1: *Good overview of the technical importance of the task*

Response: We thank the Reviewer for the positive evaluation and for recognizing the technical importance of the task addressed in this study.

R2.2: *Page 5 line 11: could benefit by specifically referencing some of the differences in longitudinal and transverse segmentations found in prior works here*

Response: We thank the Reviewer for this suggestion. We have now expanded the Introduction (Page 5) to explicitly describe the neurobiological and functional differences captured by these two segmentation axes.

"...accumulating human functional parcellation and gradient evidence highlights the importance of the longitudinal (anterior-posterior) axis. [Prior works suggest that these two axes capture distinct neurobiological dimensions: while the transverse axis reflects the intrinsic micro-circuitry and histological properties of hippocampal subfields \(e.g., the high neurogenic potential of the dentate gyrus\), the longitudinal axis manifests a functional gradient, with the anterior hippocampus primarily involved in emotional processing and the posterior segments more dedicated to cognitive functions and flexibility\(6\).](#)"

R2.3: *Page 5 line 25: write out HPA*

Response: We have written out the abbreviation as "hypothalamic-pituitary-adrenal (HPA)" axis in the revised manuscript.

R2.4: *Page 5 lines 31-34: "opposite reports remain" is awkward phrasing,*

Response: We apologize for the awkward phrasing. We have revised this sentence to: "However, conflicting findings continue to be reported."

R2.5: *Page 5 line 43: "reproducible marker" for what?*

Response: We have clarified this sentence in the revised manuscript to state: "a reproducible marker for antidepressant treatment response."

R2.6: *The last paragraph strikes me as too detailed for the intro and should be presented much more concisely; specific details including cohort size, evaluation frameworks, etc can go in methods and beginning of results, and this paragraph should just be a few sentences*

Response: We agree with the Reviewer's assessment and have significantly condensed the final paragraph of the Introduction. Specific details, including the exact cohort size (N=56 patients, 70

HCS) and the technical justification for the TabPFN classifier, have been moved to the Materials and Methods section.

R2.7: *Is there a specific hypothesis that the authors had going into the study? If so, please include*

Response: Thank you for this suggestion.

Response: We appreciate this comment. We have added a hypothesis centered on the **trade-off between anatomical granularity and signal robustness as following:**

"We hypothesized that while micro-scale subfields provide finer anatomical detail, macro-scale measures might offer more robust and clinically relevant predictive signals for treatment response due to their higher signal-to-noise ratio and greater stability under standard 3T T1-weighted imaging conditions."

This framing acknowledges the intuitive appeal of micro-scale analysis while providing a theoretical basis for our comparative investigation across different scales.

Materials and Methods:

R2.8: *Page 7 lines 13-15: DSM 5 has been out since 2013; please specify data collection ranges so readers can see how current the methodology is*

R2.9: *Page 7 lines 38-50: not sure if you need to provide this level of detail and justification in the text (as opposed to the cover letter?), as it takes up valuable space that could be used to expand on details as described in other comments; at most, you could say this in a shorter sentence*

Response: We agree and have significantly condensed this section. The justification regarding the transition from FreeSurfer v5.3 to v7.1.1 has been reduced to a single sentence focusing on the technical necessity of using the updated atlas for longitudinal hippocampal segmentation.

R2.10: *Page 10 line 20: is there a reference or justification for using these thresholds for responders and non-responders?*

Response: A $\geq 50\%$ reduction in the baseline Hamilton Depression Rating Scale (HAM-D) score is a widely accepted criterion for defining clinical response in antidepressant trials. We have now added appropriate references to support the use of this threshold.

R2.11: *Page 10 line 33: when you say "sample" does this mean only a subset of the HC group?*

Response: We apologize for the lack of clarity. "Sample" referred to the entire cohort of 70 healthy controls used to establish the normative distribution. We have changed the wording to "the entire healthy control group" to avoid any confusion.

R2.12: *Was there any subgroup analysis performed to distinguish between antidepressant classes, e.g., TCAs vs SNRIs vs SSRIs?*

Response: We thank the Reviewer for this important question. Our cohort included patients treated with SSRIs (n=34), SNRIs (n=13), and TCAs (n=9). A formal subgroup analysis for each antidepressant class was not performed, as the resulting sub-groups (particularly TCAs and SNRIs) would be severely underpowered. We addressed this potential confounder through the following statistical steps:

Second, we conducted a subgroup analysis by repeating the group comparisons (Z-scores and Asymmetry Indices) within the largest subgroup (SSRIs, n=34, Table R1). While the reduction in sample size from 56 to 34 inevitably decreased statistical power—resulting in some P-values no longer reaching significance after FDR correction—the effect sizes (η_p^2) remained robust and the directionality of findings was entirely consistent with the full-sample analysis. In our primary regions of interest, the left posterior hippocampus (body and tail), the η_p^2 in the SSRI subgroup remained between 0.08 and 0.12 for Z-scores and reached up to 0.15 for the asymmetry index of the hippocampal body, which is highly comparable to the magnitude of effects observed in the whole cohort.

Table R2. Detailed Comparison of Group Differences (RES vs. NRES) between the Full MDD Sample and the SSRI Subgroup

Hippocampal Metrics	Full MDD Sample (n=56)			SSRI Subgroup (n=34)		
	P_{unc}	η_p^2	P_{FDR}	P_{unc}	η_p^2	P_{FDR}
L-HE	0.005	0.145	0.031	0.061	0.116	0.190
L-HF	0.002	0.176	0.025	0.051	0.125	0.190
L-HP	0.010	0.123	0.031	0.066	0.112	0.190
L-Comb.Dentate	0.019	0.103	0.038	0.111	0.085	0.244
L-CA.Only	0.012	0.117	0.031	0.068	0.110	0.190
L-Comb.Dentate.CA	0.011	0.121	0.031	0.067	0.111	0.190
L-CA2.4	0.013	0.114	0.031	0.122	0.081	0.244
L-body	0.004	0.155	0.005	0.063	0.115	0.125
L-Hippocampal_tail	0.005	0.147	0.005	0.128	0.078	0.128
AI-HE	0.006	0.140	0.029	0.040	0.137	0.200
AI-HF	0.006	0.140	0.029	0.068	0.110	0.228
AI-body	0.019	0.103	0.057	0.035	0.145	0.200
AI-tail	0.028	0.091	0.057	0.221	0.051	0.230

P_{unc} , uncorrected P-value; η_p^2 , partial eta squared (representing effect size); P_{FDR} , Benjamini-Hochberg FDR-corrected P-value. The metrics listed are those that achieved significant group differentiation ($P_{FDR} < .05$) in the primary full-sample analysis.

Results

R2.13: *Would appreciate a statistical comparison of AUCs between protocols, e.g. via DeLong's test, so we can see whether these protocols truly differ in performance*

Response: We thank the Reviewer for this rigorous suggestion. We have now performed DeLong's test to find no statistically significant differences in AUCs among the three protocols. This is unsurprising because the AUCs of top winners of each protocol are very close (between 0.683 and 0.718), and support our conclusion that macro-scale measures (Regional and Composite) are **sufficient** for capturing the neuroanatomical signal, as they achieve diagnostic accuracy comparable to the more computationally intensive micro-scale subfield approach. We've mentioned this additional test in revised manuscript.

statistical comparisons of the AUCs of each protocol using determine the predictive performance differs significantly across the three protocols. We selected the top-performing ("champion")

model from each protocol: L-tail (Local), Bilateral-Body (Regional), and L-Hippocampal_Formation (L-HF, Composite). We extracted the paired risk scores for each subject from the LOOCV framework and performed the tests using the pROC package in R. The results are summarized below:

Discussion

R2.14: *Page 17 line 59: seems like some redundancy with the first paragraph of the discussion, e.g., saying macro scale features outperform micro scale features was already said*

Response: We apologize for the redundancy. We have revised the first paragraph of the discussion to more concisely summarize the findings and removed the repetitive statements. Specifically:

“..... even when latent factors were extracted via dimensionality reduction (PCA), ~~suggesting that the relevant pathological signature is captured at the regional rather than the micro structural level.~~ (2)Second, ~~based on these robust macro-scale markers, we characterized the specific neurobiological signature of treatment response.~~ we identified the left posterior hippocampus (Body and Tail)“

R2.15: *The limitation paragraph says “third” twice*

Response: Thank you for pointing out this clerical error. We have corrected the enumeration in the Limitations section.

R2.16: *I appreciate the depth of interpretation from the authors, but this seems overly interpretive given the similarity in performance between protocols and the low sample size, particularly given only 18 non-responders*

Response: We appreciate this cautionary note and agree that the modest sample size requires a more conservative interpretation. We have toned down the biological speculation in the Discussion and more explicitly framed our mechanistic hypotheses as preliminary findings that require validation in larger, multi-site cohorts. For example, in Page , line

“In line with previous studies(14, 22, 30), we found that posterior hippocampal volumes (specifically the regional body and tail) demonstrated the highest predictive efficacy ~~among the tested population.~~This restoration of flexibility effectively enables the brain to translate acute pharmacological effects into sustained symptomatic improvement. ~~However, given our modest sample size, particularly the limited number of non-responders, these interpretations remain exploratory.~~ Further research ~~with larger, multi-site cohorts~~ is required.....”

And in Page ,Line

This functional deficit in the left posterior hub mirrors the structural volumetric deviation identified in our study, ~~aligning with the possibility~~ that the left posterior hippocampus ~~may serve as a potential interface~~ for the cognitive-emotional integration required for recovery.

R2.17: *It’s valuable to make the emphasis that macro scale segmentation is sufficient, because that means the extra effort for subfield segmentation may not be worth it, but that requires a reframing of the discussion*

Response: We thank the Reviewer for this insightful suggestion, which significantly strengthens the clinical relevance of our work. We have reframed the Discussion to emphasize that macro-scale segmentation is sufficient for predicting treatment response.

For example, in the first paragraph of the discussion:

“..... we demonstrate that macro-scale measures (the composite protocols and regional) offer sufficient predictive utility and generally outperformed micro-scale local subfields.....“

And in the second paragraph of the discussion:

“Indeed, our PCA recovered which showed no predictive utility. This suggests that the variance critical for treatment prediction is effectively aggregated at the macro-scale. From a practical standpoint, this sufficiency of macro-scale markers implies that the significant computational expertise and visual segmentation quality checking often required for micro-scale subfield segmentation may not yield incremental prognostic value in clinical settings.”